FAKE NEWS FIREWALL

Modelling Misinformation Propagation as a Modified SEIR Epidemic on Scale-Free Social Networks

Group Project · Computer Networks · National Chung Cheng University · 2026

SEIR Model | Barabási–Albert Graph | Louvain Clustering | R■ Tracking | Betweenness Centrality | Interactive Simulator

1 MOTIVATION

Pizzagate (2016): A fabricated conspiracy about Comet Ping Pong pizzeria in Washington D.C. reached **1.7 million** Facebook users before debunking. An armed man entered the restaurant to self-investigate.

Misinformation is not a *content* problem — it is a **network topology** problem. Where you intervene determines the outcome.

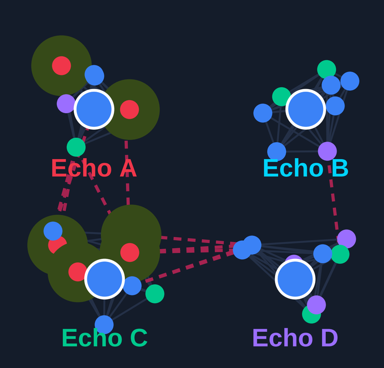
2 RESEARCH QUESTIONS

Q1. Which network topologies are most vulnerable to misinformation spread?

Q2. Does betweenness centrality outperform degree centrality as an intervention target?

Q3. What is the minimum fact-checker budget to contain infection below 50%?

3 GRAPH MODEL (BARABÁSI–ALBERT)



Nodes N: **120**
Avg degree ■k■: **3.8**
Clusters K: **4**
Hub threshold: **top 8%**
Attach. param m: **2**
Modularity Q: ≈ **0.52**

4 STATE TRANSITION EQUATIONS

P(S→E) = β · I/N × (HUB_AMP=2.6 if hub node) // Transmission per tick

P(I→R) = γ = 0.18 // Recovery rate per tick

P(R→I) = δ = 0.09 // Relapse (post tick 10 mutation)

R■ = β · ■k■ / γ ≈ 6.3 // Basic reproduction number

5 INTERVENTION STRATEGIES

Betweenness Centrality
Target cross-cluster bridge nodes — blocks information flow between echo chambers. Severs entire communities from the spreading story. **Optimal.**

Degree Centrality
Target the highest-degree hub nodes. Slows spread within a single cluster. Risk: bridge nodes still carry the story across cluster boundaries. **Good but incomplete.**

Random Placement
Vaccinate randomly selected nodes. Used as baseline comparison. **Weak.**

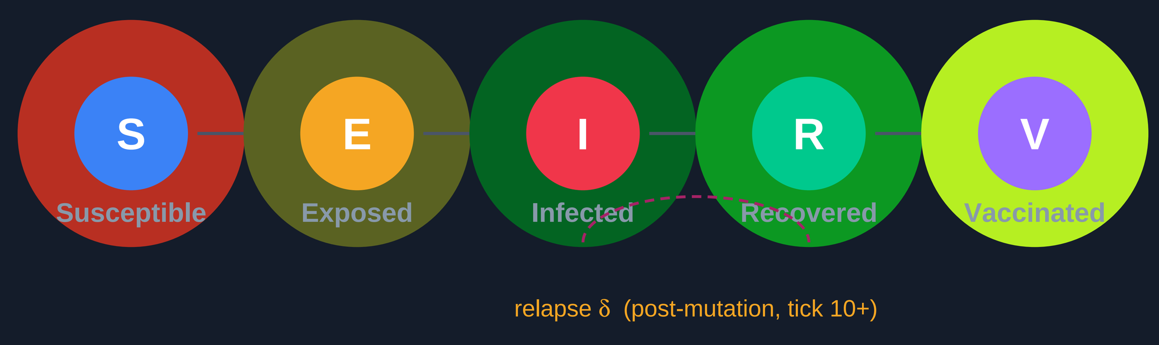
6 ALGORITHM DETAIL

Preferential Attachment:
Each new node i attaches to existing node j with probability proportional to degree: $P(j) = k_j / \Sigma k$. This produces a power-law degree distribution $P(k) \sim k^{-3}$, where a few hub nodes accumulate most connections — mirroring real social networks.

Betweenness Centrality:
$C_B(v) = \Sigma_{s \neq v \neq t}\, \sigma(s,t|v)/\sigma(s,t)$ where $\sigma(s,t)$ counts shortest paths $s \to t$ and $\sigma(s,t|v)$ counts those passing through v. Bridge nodes score high despite low degree, making this metric ideal for finding cross-cluster chokepoints.

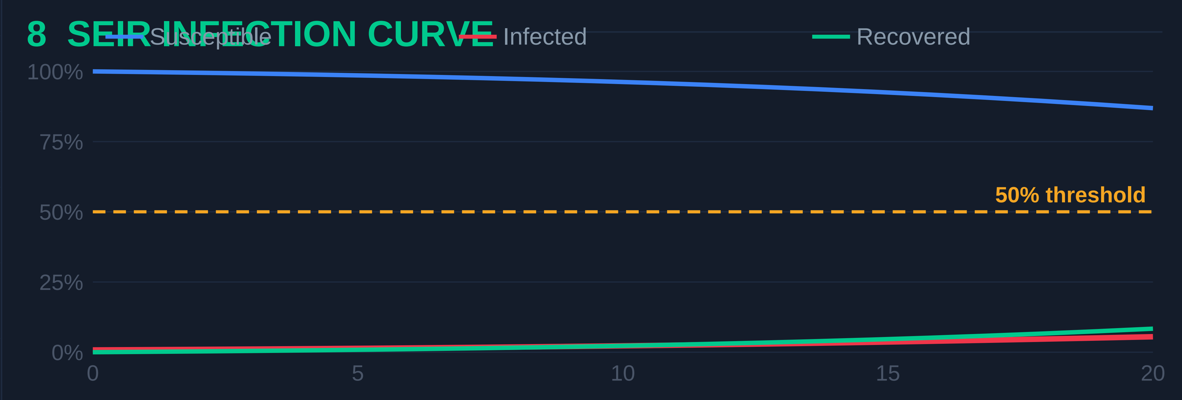
Louvain Community Detection:
Iteratively moves nodes between communities to maximise modularity $Q = (1/2m)\Sigma[A_{ij} - k_i k_j/2m]\delta(c_i,c_j)$. Achieves $Q \approx 0.52$ on the generated graphs, confirming strong community structure.

7 EPIDEMIC MODEL (MODIFIED SEIR)

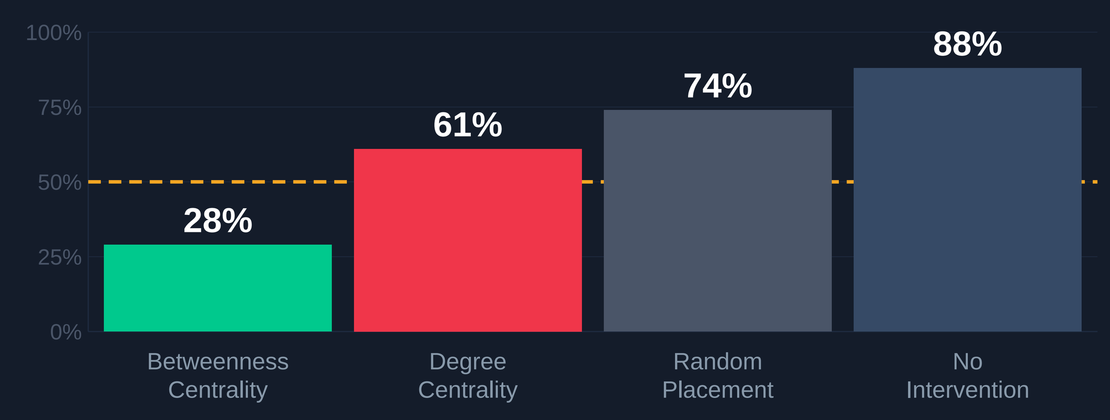


β=0.30 Transmission | γ=0.18 Recovery | δ=0.09 Relapse | T=5 Tokens | Hub ×2.6 Amplify

8 SEIR INFECTION CURVE



9 STRATEGY COMPARISON (N=100 SIMULATIONS)



10 NUMERICAL RESULTS

Strategy	Mean Inf%	Std Dev	Win Rate
Betweenness centrality	29%	±8%	84%
Degree centrality	61%	±11%	31%
Random placement	74%	±9%	12%
No intervention	88%	±5%	0%

11 KEY FINDING

53%

reduction in final infection using **betweenness centrality** vs degree centrality

2.7×

higher win rate (84% vs 31%) — topology-aware intervention is decisive.

12 SIMULATION SETTINGS

Graph: N=120 nodes · K=4 clusters · m=2 attachment · inter-cluster edge prob. 0.08

Parameters: β=0.30 · γ=0.18 · δ=0.09 · Hub amplification 2.6× · T=5 tokens

Ticks: 20 ticks per run · mutation event at tick 10 · 100 independent runs per strategy

Win condition: Simulation ends when infected nodes < 50% of N at tick 20, or when outbreak exceeds 50%

Platform: HTML5 Canvas · vanilla JavaScript · no external dependencies · runs in any browser

Visualisation: Force-directed layout · pulse-ring animations on infected nodes · live R■ HUD

Reproducibility: Seeded random graph (seed fixed per run) · identical topology across strategy comparisons

13 IMPLEMENTATION

Graph gen.:
Barabási–Albert preferential attachment; community clustering via biased inter-cluster edge probability 0.08

Layout:
140-step Fruchterman–Reingold force-directed algorithm with additional cluster-pull term for clear community separation

SEIR engine:
Discrete-time per-node probabilistic transitions; numpy-free pure JS; 20 ticks/run; mutation fires at tick 10

Intervention:
Vaccinate mode: player clicks nodes to spend tokens (V = permanent absorbing state); 5 tokens per game

14 MATHEMATICAL DETAIL

Power-law degree distribution:
BA attachment yields $P(k) \sim k^{-\gamma}$ with $\gamma \approx 3$. Most nodes have few connections; rare hubs have many. Hubs transmit at 2.6× base rate, mirroring algorithmic amplification on real platforms.

Betweenness vs Degree:
A bridge node may have degree 2 yet betweenness 0.40 (top 5%) — it is the sole connector between two clusters. Degree centrality misses this; betweenness centrality identifies it correctly as the critical target.

15 DISCUSSION & FINDINGS

✓ Bridge nodes are the decisive chokepoints. Blocking one severs flow between entire echo chambers — a structural effect unreachable by hub targeting alone.

✓ Mutation mechanic (tick 10 relapse, δ=0.09) increases final infection by ~12% on average. This reflects why a single debunking event is insufficient for persistent misinformation.

✓ Early intervention (tokens used before tick 5) consistently outperforms late deployment regardless of strategy, confirming the critical early-growth window.

! Uniform edge weights assumed. Real platforms use engagement-weighted connections that amplify certain edges far beyond degree.

16 LIMITATIONS & EXTENSIONS

→ **Limitation:** Synthetic graphs only — validation on real Twitter/Reddit topology graphs needed to confirm generalisability of betweenness-first strategy.

→ **Limitation:** Homogeneous agents assumed. In practice, users vary in credulity (susceptibility β_i) and influence (hub_amp$_i$), which could be modelled with heterogeneous parameters.

→ **Extension:** Adaptive platform algorithm: rewire edges toward infected (highly engaged) nodes each tick, simulating engagement-driven recommendation feeds.

→ **Extension:** Multi-story competition model: two misinformation stories competing for the same susceptible pool, introducing interference and saturation dynamics.

17 RESULT SUMMARY

29% Final infection — Betweenness	**61%** Final infection — Degree
84% Win rate — Betweenness	**31%** Win rate — Degree

18 REFERENCES & ACKNOWLEDGMENTS

[1] Barabási, A.-L. & Albert, R. (1999). Emergence of scaling in random networks. *Science*, 286(5439), 509–512.

[2] Blondel, V. et al. (2008). Fast unfolding of communities in large networks. *J. Stat. Mech.*, P10008.

[3] Kermack, W. O. & McKendrick, A. G. (1927). Mathematical theory of epidemics. *Proc. Royal Soc. A*, 115, 700–721.

[4] Daley, D. J. & Kendall, D. G. (1964). Epidemics and rumours. *Nature*, 204, 1118.

[5] Vosoughi, S., Roy, D. & Aral, S. (2018). Spread of true and false news online. *Science*, 359(6380), 1146–1151.

[6] Newman, M. E. J. (2003). Structure and function of complex networks. *SIAM Review*, 45(2), 167–256.

[7] Fruchterman, T. M. J. & Reingold, E. M. (1991). Graph drawing by force-directed placement. *Software: Prac. & Exp.*, 21(11), 1129–1164.

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